

SLIIT Faculty of Computing

Year 2 Semester 1 - 2026

Individual Machine Learning Report

IT2011 - Artificial Intelligence and Machine Learning

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Assigned model: Multinomial Naive Bayes

Primary preprocessing specialization: Feature selection

Emotion-Aware Movie Recommendation System

Final Report

Executive Summary

This report presents an individual implementation of Multinomial Naive Bayes for seven-class emotion prediction from movie descriptions. The work independently demonstrates missing-data handling, categorical encoding, outlier treatment, normalization/scaling, feature engineering, feature selection, dimensionality reduction, exploratory analysis, tuning, cross-validation, and held-out evaluation. The achieved test macro F1 was **0.2362** and test accuracy was **0.3155**.

1. Introduction and Problem Statement

Conventional movie recommenders emphasize popularity, ratings, or viewing history and may not represent the emotional character of movie content. This project predicts one of seven emotions - anger, anticipation, disgust, fear, joy, optimism, or sadness - from a movie description and combines the result with genre and rating information for emotion-aware recommendation.

2. Dataset Description

The assigned IMDb Movie Reviews - Genres, Descriptions and Emotions CSV originally contained 46,173 rows and eight columns. Description-level aggregation produced 2,060 independent descriptions for EDA and 2,057 records for seven-class modelling. The surprise class was retained for EDA but excluded from training because only three independent examples were available.

Characteristic	Value
Raw observations	46,173
Full EDA observations	2,060
Seven-class observations	2,057
Target	emotion
Primary text feature	Description
Supporting features	Genre, rating, length, popularity
Missing processed values	0

3. Data Preprocessing and Exploratory Data Analysis

The primary specialization was **Feature selection**. The complete workflow still independently demonstrated every compulsory preprocessing technique. All learned transformations used during modelling were placed inside a pipeline and fitted on training folds only.

Technique	Implementation and justification
Missing data	Audited nulls and blanks; represented unavailable genre information as unknown_genre; used fold-fitted imputers.
Categorical encoding	One-hot encoded genre categories with unknown-category handling; demonstrated target label encoding.
Outliers	Detected numeric extremes using the 1.5 IQR rule and clipped them instead of deleting valid minority observations.
Scaling	Scaled numeric inputs; preserved non-negative values before chi-squared selection and standardized dense neural input where applicable.
Feature engineering	Created rating spread, text length, genre count, review popularity, and log-count features.
Feature selection	Used training-fold chi-squared selection to retain target-associated non-negative features.
Dimensionality reduction	Demonstrated TruncatedSVD; used it in dense-input models where appropriate.

EDA Visualizations and Interpretation

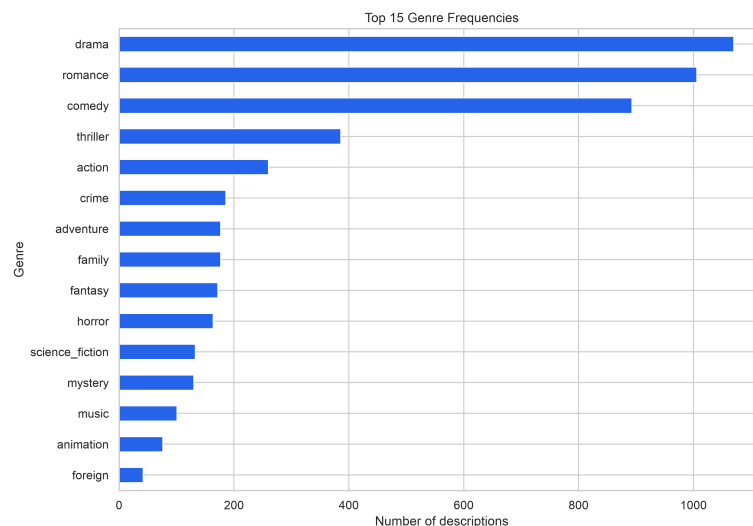


Figure 1. A few genres dominate, while rare genres offer limited class evidence.

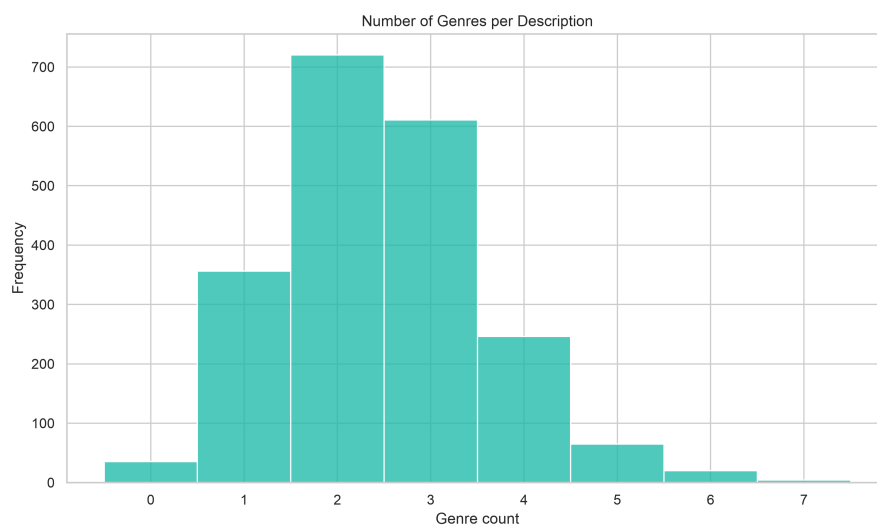


Figure 2. Most descriptions contain a small number of genres; unavailable genres use an explicit category.

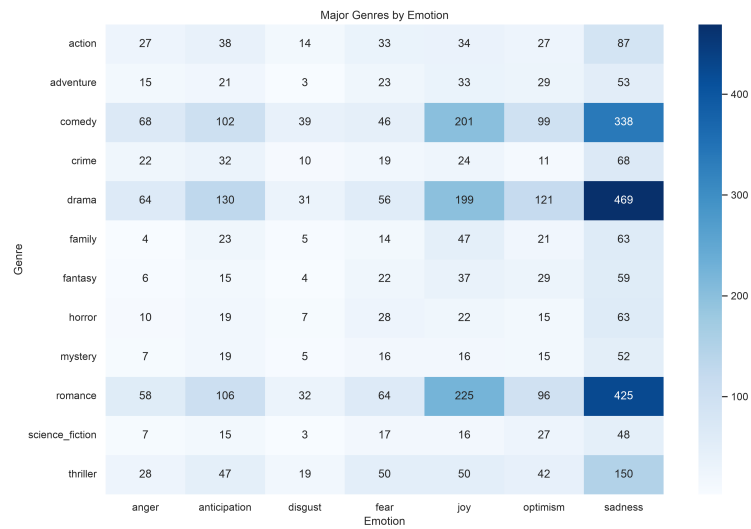


Figure 3. Genre-emotion frequencies vary but are also influenced by overall class imbalance.

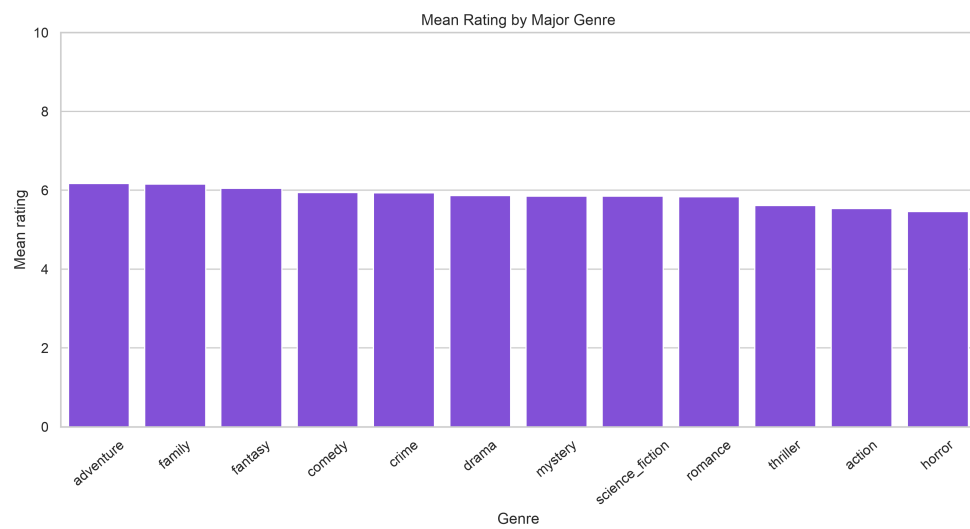


Figure 4. Ratings differ modestly by genre and are insufficient as standalone predictors.

4. Model Design and Implementation

Suitability. Multinomial Naive Bayes is an efficient probabilistic baseline for non-negative TF-IDF and encoded features.

The implementation used pandas, NumPy, Matplotlib, Seaborn, and scikit-learn. A stratified 80:20 split with random state 42 preserved class proportions. GridSearchCV used five-fold StratifiedKFold and optimized macro F1. The final test set remained untouched during tuning.

Hyperparameter varieties

Alpha = 0.1, 0.5, 1.0, 2.0; fitted/uniform priors; 500, 1,000, and 1,500 selected features; unigram/bigram TF-IDF.

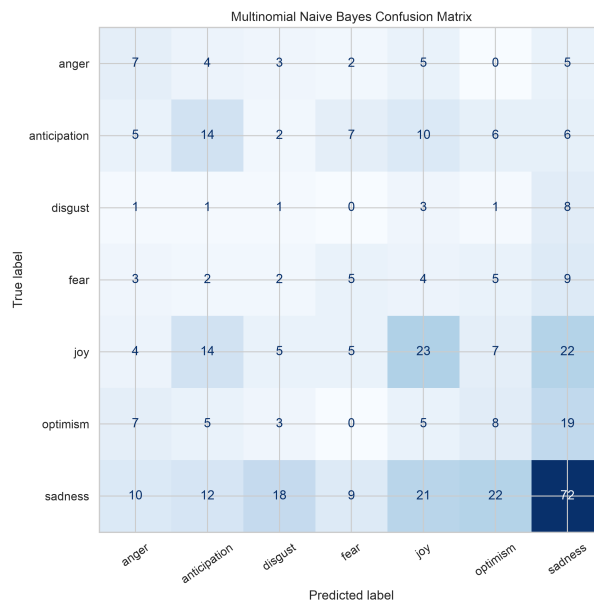
Selected parameters

'model__alpha': 0.1, 'model__fit_prior': False, 'preprocess__text__tfidf__ngram_range': [1, 1], 'select__k': 1500

5. Model Evaluation and Comparison

Metric	Score
Training accuracy	0.8553
Test accuracy	0.3155
Macro precision	0.2378
Macro recall	0.2399
Macro F1	0.2362
Weighted F1	0.3275
Best CV macro F1	0.2651

Macro F1 was emphasized because it gives equal importance to every class. Weighted F1 and accuracy are also reported but are influenced more strongly by the dominant sadness and joy classes.



Confusion matrix for the selected tuned variety.

Group-level comparison

Rank	Model	Accuracy	Macro F1	Weighted F1	CV Macro F1
1	Linear SVM	0.3641	0.2523	0.3661	0.2623
2	Logistic Regression	0.3155	0.2387	0.3322	0.2662
3	Multinomial Naive Bayes	0.3155	0.2362	0.3275	0.2651
4	Multi-Layer Perceptron	0.3592	0.1916	0.3297	0.2068
5	Random Forest	0.2791	0.1677	0.2751	0.2296
6	Decision Tree	0.2621	0.1523	0.2565	0.1772

For this model, the training-test accuracy gap was 0.5398. A large gap indicates overfitting; similarly weak training and test performance may indicate underfitting. Class-level results showed that disgust remained especially difficult because only 73 independent examples were available.

6. Ethical Considerations and Bias Mitigation

The class imbalance creates unequal error risks and may underrepresent minority emotions. Emotion labels are subjective, descriptions may contain cultural or linguistic bias, and a low-confidence prediction should not be treated as a psychological assessment. Mitigation included macro-F1 reporting, stratification, balanced-class tuning where supported, class-level evaluation, transparent limitations, and retaining no personal user information. A production system should add confidence calibration, abstention, user feedback, and fairness monitoring.

7. AI Tool Usage Declaration and Transparency

Tool	Use	Extent	Verification
ChatGPT / Codex	Code structure, debugging suggestions, documentation organization, explanation and grammar refinement	Moderate	All notebooks were executed; metrics, artifacts, model reloads, and outputs were checked. The student remains responsible for understanding and explaining every cell.

8. Individual Reflections and Lessons Learned

The work showed that preprocessing must be justified by the data and kept inside training folds. The primary responsibility, feature selection, was not an isolated step: it interacted with feature representation, computational cost, and generalization. The final results also showed why macro F1, class-level recall, and overfitting analysis are necessary instead of relying only on accuracy.

9. References

1. Fahad Rehman. IMDb Movie Reviews - Genres, Descriptions and Emotions. Kaggle.
2. Pedregosa et al. Scikit-learn: Machine Learning in Python. JMLR, 2011.
3. Salton and Buckley. Term-weighting approaches in automatic text retrieval, 1988.
4. SLIIT IT2011 Assignment Specification and Final Evaluation Guidelines, 2026.
5. Project notebooks, dataset audit, data dictionary, and executed result files.